

PEAK ACHIEVERS
ONLINE ACADEMY

Empowering Learners. Building Greater Futures.

GRADE
10

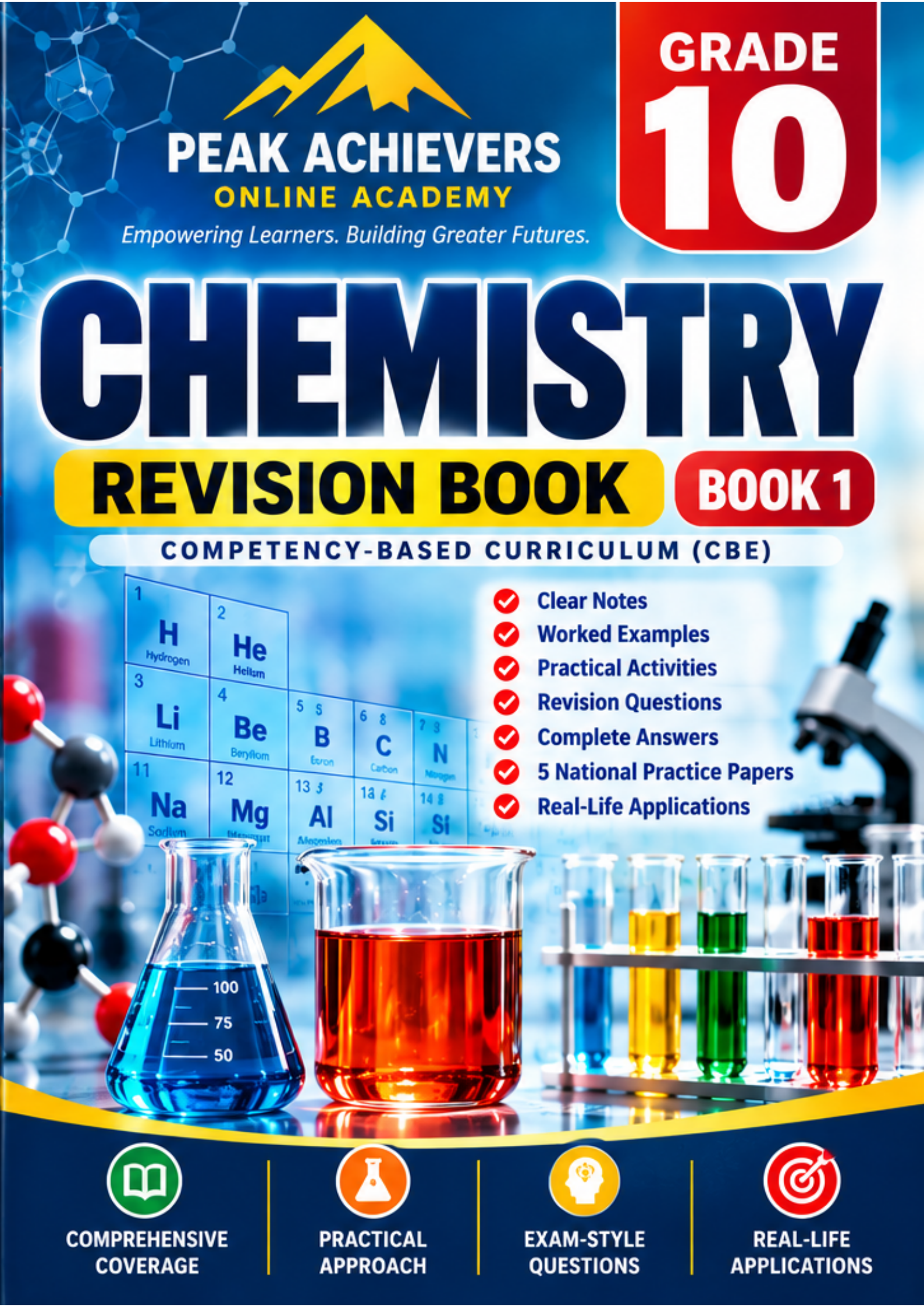
CHEMISTRY

REVISION BOOK

BOOK 1

COMPETENCY-BASED CURRICULUM (CBE)

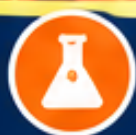
- ✓ Clear Notes
- ✓ Worked Examples
- ✓ Practical Activities
- ✓ Revision Questions
- ✓ Complete Answers
- ✓ 5 National Practice Papers
- ✓ Real-Life Applications



1 H Hydrogen	2 He Helium			
3 Li Lithium	4 Be Beryllium	5 B Boron	6 C Carbon	7 N Nitrogen
11 Na Sodium	12 Mg Magnesium	13 Al Aluminum	14 Si Silicon	15 P Phosphorus



**COMPREHENSIVE
COVERAGE**



**PRACTICAL
APPROACH**



**EXAM-STYLE
QUESTIONS**



**REAL-LIFE
APPLICATIONS**

PEAK ACHIEVERS ONLINE ACADEMY

SENIOR SCHOOL CHEMISTRY

GRADE 10

Competency-Based Education (CBE)

BOOK 1 • FIRST EDITION 2026

Prepared by

Hezekiah Omweri Nyatuka

B.Ed. (Science), Kenyatta University

Chemistry & Physics

Learn • Apply • Achieve

Copyright & Disclaimer

Copyright © 2026 Hezekiah Omweri Nyatuka. First Edition, 2026. Published under Peak Achievers Online Academy. All rights reserved.

This revision book is an independently prepared educational resource for Grade 10 Senior School Chemistry. The practice assessments are original learning materials and are not official examination papers issued by KNEC or any curriculum authority.

About the Author

Hezekiah Omweri Nyatuka is a Bachelor of Education (Science) graduate of Kenyatta University, specialising in Chemistry and Physics. Through classroom teaching, tutoring and Peak Achievers Online Academy, he supports learners with structured revision, practical scientific reasoning and competency-based learning.

Dedication

This book is dedicated to the loving memory of my late father, Simeon Nyatuka, whose guidance, sacrifice, and belief in education continue to inspire my journey and the work I do as an educator.

To my mother, Hyrine Kerubo, for her love, strength and unwavering support throughout my life.

To my guardians, Peter Munde Mong'are and Teresa Kwamboka Sawe, for their guidance, encouragement and support.

To my friends, Mnaree, Richie, Duke, Caleb, Cuba, Philo, Rob, Tim, Mr Peter Kimani, and many others, whose friendship and encouragement have been part of my journey.

To Her, my special one, for being a special part of my journey and for the place you hold in my heart.

To my fellow teachers, whose commitment to education continues to inspire the work of helping learners succeed.

And to every learner who opens this book: may your curiosity grow, your confidence strengthen and your effort lead you towards excellence.

To each of you, this book carries a part of your contribution to my journey.

— Hezekiah Omweri Nyatuka

Preface

This book has been developed to support Grade 10 Senior School Chemistry learners through concise explanations, carefully selected examples, practical activities, application questions and competency-based revision exercises. The emphasis is on understanding, investigation, evidence, problem solving and application rather than memorisation alone.

How to Use This Book

- Read the brief notes carefully and study the equations, tables and diagrams.
- Use Remember, Note, NB, Hint and Exam Focus boxes to identify key ideas and common errors.
- Complete practical and application activities using evidence-based scientific reasoning.
- Attempt revision questions before consulting answers and marking guidance.
- For calculations use: Formula → Substitution → Calculation → Answer.
- For investigations use: Observation → Interpretation → Conclusion.

Competency Development

The book supports communication and collaboration, critical thinking and problem solving, creativity and imagination, digital literacy, learning to learn, citizenship and self-efficacy through chemistry contexts.

Laboratory Safety

- Follow the teacher's or laboratory supervisor's instructions.
- Wear appropriate protective equipment.
- Never taste laboratory chemicals.
- Handle acids, alkalis and heated apparatus carefully.
- Report spills, broken apparatus and injuries immediately.
- Dispose of chemicals according to laboratory instructions.

CONTENTS

- Chapter 1 — Introduction to Chemistry
- Chapter 2 — The Atom
- Chapter 3 — The Periodic Table
- Chapter 4 — Chemical Bonding
- Chapter 5 — Periodicity
- Chapter 6 — Acids and Bases

CHAPTER 1

INTRODUCTION TO CHEMISTRY

Learning Outcomes

- Explain the meaning of Chemistry.
- Identify and describe major branches of Chemistry.
- Appreciate applications of Chemistry in everyday life.
- Identify Chemistry-related careers.
- Explain responsible use of drugs and chemical substances.
- Demonstrate safe and healthy Chemistry learning practices.

1.1 Meaning of Chemistry

Chemistry is the branch of science that studies matter, its composition, structure, properties and the changes it undergoes.

Matter is anything that has mass and occupies space.

1.2 Major Branches of Chemistry

Analytical Chemistry — identification and measurement of substances.

Biochemistry — chemical processes in living organism.

Physical Chemistry — energy and physical principles involved in chemical processes.

Inorganic Chemistry — elements and compounds other than most carbon compounds.

Organic Chemistry — carbon-containing compounds. NOTE: introduced here only as a branch; it is not studied as a topic in this Grade 10 Book.

QUICK CHECK

Which branch mainly identifies and measures substances? Analytical Chemistry.

1.3 Chemistry in Everyday Life

Chemistry is applied in agriculture, medicine, food, water treatment, construction, cleaning, environmental protection and materials.

REMEMBER: Chemistry is not limited to the laboratory.

1.4 Chemistry-Related Careers

Examples include chemist, laboratory technologist, pharmacist, chemical engineer, food scientist, environmental scientist, teacher, agricultural scientist, forensic scientist and quality-control analyst.

1.5 Responsible Use of Drugs and Chemical Substances

Read labels; use substances only as intended; never taste chemicals; avoid deliberate inhalation; do not mix chemicals unless instructed; store safely; use medicines according to professional or label instructions; dispose of chemicals safely.

CHAPTER 2

THE ATOM

Learning Outcomes

- Describe the basic structure of an atom.
- Identify protons, neutrons and electrons.
- Use atomic number and mass number.
- Explain isotopes.
- Write simple electron arrangements.
- Calculate relative atomic mass from simple isotope data.

Initial Checkpoint

1. What is an atom? 2. Name the three main subatomic particles. 3. State the relative charge of each. 4. Where are protons and neutrons found? 5. Where are electrons found? 6. What is atomic number? 7. What is mass number? 8. What are isotopes?

2.1 Structure of the Atom

An atom is the smallest particle of an element that takes part in a chemical reaction while retaining the chemical identity of that element.

Proton: charge +1, relative mass 1, nucleus. Neutron: charge 0, relative mass 1, nucleus. Electron: charge -1, relative mass approximately $1/1840$, outside the nucleus in energy levels.

Figure 2.1 Structure of the atom — nucleus containing protons and neutrons, with electrons arranged in energy levels.

Particle	Relative charge	Relative mass	Location
Proton	+1	1	Nucleus
Neutron	0	1	Nucleus
Electron	-1	$\approx 1/1840$	Energy levels outside nucleus

2.2 Atomic Number and Mass Number

Atomic number, Z = number of protons. Mass number, A = protons + neutrons. Therefore neutrons = $A - Z$. In a neutral atom, protons = electrons.

Example: for $Z = 11$ and $A = 23$: protons = 11, electrons = 11, neutrons = $23 - 11 = 12$.

2.3 Nuclide Notation

Nuclide notation is written as ${}^A_Z\text{X}$, for example ${}^{23}_{11}\text{Na}$.

2.4 Development of the Atomic Model

Dalton described the atom as a small solid particle. Rutherford proposed a small dense positively charged nucleus with electrons outside it and showed that much of the atom is empty space.

THINK BIG: Scientific models change when new evidence becomes available.

2.5 Isotopes

Isotopes are atoms of the same element with the same number of protons but different numbers of neutrons.

2.6 Relative Atomic Mass

$A_r = \Sigma(\text{isotopic mass} \times \text{percentage abundance}) / 100$. Example: 75% chlorine-35 and 25% chlorine-37 gives $A_r = 35.5$. Relative atomic mass has no unit.

2.7 Electron Arrangement

Examples: H 1; He 2; Li 2,1; C 2,4; O 2,6; Ne 2,8; Na 2,8,1; Cl 2,8,7; Ca 2,8,8,2.

Element	Atomic number	Electron arrangement
O	8	2,6
Na	11	2,8,1
Cl	17	2,8,7
Ca	20	2,8,8,2

TOPICAL PRACTICE

1. Complete a table for proton, neutron and electron showing charge, relative mass and location.
2. For $Z = 11$ and $A = 23$ determine protons, electrons and neutrons.
3. Write nuclide notation for sodium-23.
4. State one idea associated with Dalton's model and one with Rutherford's model.
5. Chlorine-35 and chlorine-37 are isotopes. Determine their neutron numbers and explain why they are isotopes.
6. Calculate A_r for 75% X-35 and 25% X-37.
7. Write electron arrangements for O, Na, Cl and Ca.

COMMON LEARNER ERRORS: atomic number is not neutron number; mass number is not neutron number; isotopes have the same proton number but different neutron numbers; A_r has no unit.

Essential Reference

Atomic number = number of protons.

Mass number = protons + neutrons.

For a neutral atom: number of protons = number of electrons.

Isotopes are atoms of the same element with the same number of protons but different numbers of neutrons.

CHAPTER 4

CHEMICAL BONDING

Learning Outcomes

- Explain ionic, covalent and metallic bonding.
- Use electron arrangements to represent bond formation.
- Relate structure and bonding to properties.
- Compare ionic, covalent and metallic substances.
- Apply bonding ideas to everyday materials.

4.1 Why Atoms Form Bonds

Atoms form bonds to achieve more stable electron arrangements. Explanations should focus on outer electrons.

4.2 Ionic Bonding

Electron transfer forms positive and negative ions. The ionic bond itself is the strong electrostatic attraction between oppositely charged ions.

4.3 Properties of Ionic Compounds

Connect melting point and electrical conductivity to the ionic structure and the mobility of ions.

4.4 Covalent Bonding

Covalent bonding involves sharing electron pairs between non-metal atoms.

4.5 Metallic Bonding

Metallic bonding is the electrostatic attraction between positive metal ions and delocalised electrons.

4.6 Revision Questions

REVISION & APPLICATION

1. Define a chemical bond.
2. Distinguish ionic and covalent bonding in terms of electrons.
3. Using sodium and chlorine, explain ion formation, ionic bonding and the formula of sodium chloride.
4. Explain the formation of magnesium oxide from magnesium and oxygen atoms.
5. Explain why an ionic compound does not conduct when solid but conducts when molten.
6. State the bond type in H_2 , O_2 and N_2 .
7. Explain why copper conducts electricity using metallic bonding.
8. Compare ionic, covalent and metallic bonding in terms of outer electrons.

CHAPTER 5

PERIODICITY

Learning Outcomes

- Explain the meaning of periodicity.
- Relate electron arrangement to recurring chemical behaviour.
- Describe relevant trends down Group 1 and Group 17.
- Use displacement evidence to compare reactivity.
- Apply periodic trends to simple Grade 10 predictions.

5.1 Core Grade 10 Periodicity

- Meaning of periodicity and relation to electron arrangement.
- Simple atomic-size reasoning: down a group, atomic size generally increases as occupied energy levels increase; across a period, atomic size generally decreases as nuclear charge increases while electrons are added to the same principal energy level.
- Group 1 reactivity trend and explanation using energy levels, shielding and electron loss.
- Group 17 reactivity trend and halogen displacement evidence.
- Observation → Interpretation → Conclusion in displacement investigations.

5.2 Atomic Size

Down a group, atomic size increases because additional occupied energy levels are added. Across a period, atomic size generally decreases because nuclear charge increases while electrons are added to the same main energy level.

5.3 Group 1

Group 1 elements have one outer electron. Down the group: more energy levels → greater shielding → weaker attraction for the outer electron → easier electron loss → increased reactivity.

Figure 5.1 Comparison of Group 1 and Group 17 trends.

Trend down group	Group 1	Group 17
Atomic size	Increases	Increases
Shielding	Increases	Increases
Reactivity	Increases	Decreases
Key reason	Outer electron lost more easily	Incoming electron attracted less strongly

5.4 Group 17

Group 17 elements have seven outer electrons and tend to gain one electron. Reactivity decreases down the group: $\text{Cl}_2 > \text{Br}_2 > \text{I}_2$.

CHAPTER 6

ACIDS AND BASES

Learning Outcomes

- Distinguish acids, bases and alkalis.
- Use indicators and pH to classify solutions.
- Explain neutralisation.
- Describe reactions of acids with metals, carbonates and metal oxides.
- Apply acid-base chemistry to familiar contexts.

6.1 Acids, Bases and Alkalis

An acid produces $\text{H}^+(\text{aq})$ in aqueous solution. A base reacts with an acid to form salt and water. An alkali is a soluble base that produces $\text{OH}^-(\text{aq})$ in water.

All alkalis are bases, but not all bases are alkalis.

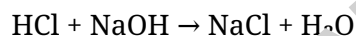
6.2 Indicators and pH

pH < 7: acidic; pH = 7: neutral; pH > 7: alkaline. Strength and concentration are not the same idea.

Figure 6.1 Simplified pH scale.

pH range	Classification	Interpretation
0–6	Acidic	pH below 7
7	Neutral	Neither acidic nor alkaline
8–14	Alkaline	pH above 7

6.3 Neutralisation



6.4 Common Acid Reactions

- $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$. Hydrogen gives a squeaky pop with a lighted splint.

Figure 6.2 Evidence used to identify common gases in this chapter.

Gas	Test	Positive observation
Hydrogen	Apply a lighted splint	Squeaky pop
Carbon dioxide	Bubble through limewater	Limewater turns milky

- $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$. Carbon dioxide turns limewater milky.
- $\text{CuO} + \text{H}_2\text{SO}_4 \rightarrow \text{CuSO}_4 + \text{H}_2\text{O}$.

6.5 Salt Naming from Acids

Hydrochloric acid → chlorides; sulfuric acid → sulfates; nitric acid → nitrates.

CHAPTER 7

INTRODUCTION TO SALTS

Learning Outcomes

- Define and name common salts.
- Distinguish soluble and insoluble salts.
- Select an appropriate salt-preparation method.
- Explain crystallisation and precipitation.
- Apply filtration, washing and drying correctly.

7.1 Salt Formation and Naming

A salt is an ionic compound formed when replaceable hydrogen ion(s) of an acid are replaced by suitable positive ions.

7.2 Choosing a Preparation Method

Soluble salt → suitable reaction followed by solution recovery/crystallisation. Insoluble salt → precipitation.

7.3 Copper(II) Sulfate from Copper(II) Oxide

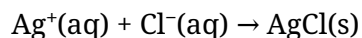
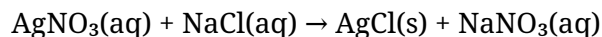
Warm dilute sulfuric acid → add CuO to excess – filter → concentrate filtrate → cool → crystallise → collect and dry crystals.

Figure 7.2 Copper(II) sulfate preparation sequence.

Step	Purpose
Warm dilute H ₂ SO ₄	Helps the reaction proceed
Add CuO to excess	Ensures acid reacts completely
Filter	Removes excess CuO
Concentrate filtrate	Removes some water
Cool	Allows crystals to form
Collect and dry	Obtains the salt crystals

Do not simply evaporate the solution to dryness.

7.4 Precipitation



Mix suitable solutions → form precipitate → filter → wash → dry.

PART IV — MIXED REVISION & ASSESSMENT

Mixed Revision 1 — Foundation & Connection

28 questions; Chapters 1–7; foundation links across laboratory skills, atomic structure, periodicity, bonding, acids/bases and salts.

Questions

1. Define Chemistry and state two everyday applications. [3]
2. State the branches concerned with identifying/measuring substances and with chemical processes in living organisms. [2]
3. An unlabelled colourless liquid is found in the laboratory. State two unsafe actions to avoid and the correct action. [3]
4. Name suitable apparatus for measuring volume, temperature and filtration, and state one use of each. [6]
5. Complete a proton/neutron/electron table showing charge, relative mass and location. [6]
6. For $^{27}_{13}\text{X}$ determine protons, neutrons, electrons and electron arrangement. [4]
7. Chlorine is 75% chlorine-35 and 25% chlorine-37. Calculate Ar. [4]
8. Explain why chlorine-35 and chlorine-37 are isotopes. [2]
9. For electron arrangement 2,8,1 state atomic number, period and group. [3]
10. For 2,8,7 state period, group, likely ion charge and explain. [4]
11. Write formulae of magnesium chloride, aluminium oxide and calcium nitrate. [3]
12. Balance $\text{Al} + \text{O}_2 \rightarrow \text{Al}_2\text{O}_3$. [2]
13. Distinguish ionic and covalent bonding. [4]
14. Explain formation of sodium chloride from 2,8,1 and 2,8,7. [5]
15. Explain why sodium chloride does not conduct when solid but conducts when molten. [4]
16. State the shared electron pairs and bond types in H_2 , O_2 and N_2 . [6]
17. Define periodicity. [2]
18. State and explain atomic-size trends down a group and across a period. [6]
19. Explain why Group 1 reactivity increases down the group. [4]
20. Chlorine is added to potassium bromide. Predict, write the equation and infer relative reactivity. [4]
21. Distinguish acid, base and alkali. [3]
22. Classify pH 3, 7 and 11; identify the alkali. [4]
23. For $\text{Mg} + \text{dilute HCl}$, write the equation, name the gas and state its test. [5]
24. A carbonate reacts with dilute HCl and bubbles. State what bubbling shows and how to identify CO_2 . [4]

PEAK ACHIEVERS ONLINE ACADEMY

GRADE 10 CHEMISTRY - FREE SAMPLE

Learn • Revise • Excel

Continue with the Complete Book

Book 1 - Revision Book - KSh 350

Get the complete Grade 10 Chemistry resource with all chapters, revision/application tasks, practical reasoning and assessment support.

Bundle: Book 1 + Book 2 - KSh 600

M-PESA Send Money: 0740 355 410

WhatsApp / inquiries: 0740 166 105 or 0740 355 410

Email: hezekiahomweri@gmail.com

PEAK ACHIEVERS ONLINE ACADEMY

Independent educational resource. Practice materials are not official KNEC papers.